

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 02-302-1
LOW VOLTAGE AUTOMATIC POWER FACTOR CORRECTION	Date:05-08-2025

EDMS 02-302-1

TECHNICAL SPECIFICATION

FOR

LOW VOLTAGE AUTOMATIC POWER FACTOR

CORRECTION PANEL (APFC)

Issue: Aug. 2025 / Rev- ١

- **This revision contains option items that must be selected by EDC before bidding.**

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EEHC DISTRIBUTION MATERIALS SPECIFICATION**EDMS 02-302-1****LOW VOLTAGE AUTOMATIC POWER FACTOR CORRECTION****Date:05-08-2025****1. SCOPE**

This specification covers the minimum technical requirements for Design, Engineering, Manufacture, testing at the manufacturer's factory, Site testing and Commissioning of Automatic Power Factor Correction (APFC) panels intended for use with low-voltage equipment to improve power factor and /or mitigate harmonics in AC electrical power systems, this specification is in addition to the special requirements of the Distribution Company.

2. APPLICABLE STANDARDS

This specification shall be read in conjunction with latest revisions of the following EDMS specifications:

Table (1)

No.	Standard No.	Description
1	EDMS19-303	Low voltage circuit breakers
2	EDMS19-300	L.V H.R.C. Fuses LINK
3	EDMS19-301	L.V H.R.C. Fuses BASE
4	EDMS17-300	Low voltage current transformers
5	EDMS21-301	Low voltage capacitor
6	EDMS06-300	TECHNICAL SPECIFICATION FOR SINGLE CORE PVC-INSULATED FLEXIBLE NON SHEATHED CABLES (WIRES)

Unless otherwise specified in this specification should be comply with the latest edition of IEC standard and should be designed, manufactured and tested in accordance to the latest applicable IEC standards as follows:

No.	Standard No.	Description
1	IEC 61921	Power capacitors-low voltage power factor correction banks
2	IEC 60831-1&2	Capacitor units and capacitor banks intended to be used, Particularly, for power-factor correction of A.C power Systems having a rated voltage up to and including 1000 V and frequencies 50 Hz. Guidelines for performance, rating and testing. Safety requirements, Guide for installation and operation.
3	IEC 60529	Classification of degree of protection provided by enclosures.
4	IEC 61557-1	Electrical safety in low voltage distribution systems up to 1000VAC.
5	IEC 60947-1	Low-voltage switchgear and control gear. part 1 :general rules
6	IEC 60947-2	Low-voltage switchgear and control gear. Part 2 : circuit breakers
7	IEC 60947-4-1	Low-voltage switchgear and control gear. Part 4-1 : contactors and motor starters.
8	ISO 2063	Metallic coatings-protection of iron and steel against corrosion- metal Spraying of zinc and aluminum.
9	IEC 60439-1	Low-voltage switchgear and control gear assemblies-type tested and Partially type tested assemblies.
10	IEEE 18	Standard for Shunt Capacitor.
11	IEEE 519	Application of Harmonic Limits.

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12	IEC 60076-6	Power Transformer reactor.
13	IEC 60085-1	Electrical insulation – thermal evaluation and designation
14	IEC 60664-1	Insulation coordination for equipment within low voltage systems
15	IEC 61326	Electrical equipment for measurement, control and laboratory use – EMC requirements
16	IEC 61439-1	Low voltage switchgear and control gear assemblies part 1:general rules
17	IEC 62208	Empty enclosures for low voltage switchgear and control gear assemblies – general requirements
18	IEC 62262	Degrees of protection provided by enclosures for electrical equipment against external mechanical impacts (IK code)
19	IEC 61000-6-4	Electromagnetic compatibility – generic standards – emission standard for industrial environments
20	IEC 61000-6-2	Generic standards – immunity for industrial environments
21	IEC61642	Industrial a.c. networks affected by harmonics-application of filters and shunt capacitors

3. ENVIRONMENTAL CONDITIONS

The electrical and mechanical properties of the required APFC panel should be guaranteed under the following environmental conditions:

Table (2)

Annual average ambient temperature	35 °C
Ambient temperature	-5°C to +55°C
Average relative humidity	95 % over a period of 24 h and 90 % over a period of one month.
Altitude	Up to 1000 m above sea-level

4. GENERAL TECHNICAL DATA

The required APFC panel should have a safe behavior and meet all requirements given below:

Table (3)

System voltage	400 V
Nominal operating (Service) voltage	380 V
Rated frequency	50 Hz.
IP class for Enclosure	IP 41 (for indoor) IP 54 (for outdoor)

5. CONSTRUCTION REQUIREMENTS

- The Design and Component Selection** of the Automatic Power Factor Correction (APFC) system shall consider continuous operation at maximum withstand voltage of 440/480/525 V(as option) and ensure reliable performance in consideration of room ambient temperature of 50°C.
- The APFC components shall be of Type Tested according to IEC relevant.
- Component Withstand Capability:** All components shall be able to withstand dynamic, thermal, and dielectric stresses resulting from prospective short-circuit currents without damage or injury to personnel.

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4. **Switching Device Requirement:** all capacitor banks shall be provided with a switching device equipped with an automatic trip.
5. **Harmonic Current Consideration:** Due to the presence of harmonic currents and manufacturing tolerances, components (MCCB, BUSBAR, and POWER CABLES) shall be designed for 1.5 times the nominal current according to the EGYPTAIN CODE.
6. **Mounting Arrangement of Capacitors:** shall allow for visual inspection and maintenance access while allowing an individual capacitor to be removed without disconnection or removal of any adjacent capacitors.
7. **Capacitor Mounting Flexibility:** Power capacitors shall also be capable of being mounted in both vertical and horizontal positions.
8. **Installation Location:** The APFC equipment shall be installed in cool, ventilated locations away from other heat-radiating elements.
9. **Protection Against Environmental Factors:** Adequate protection shall be provided against dust and direct solar radiation.
10. **Ventilation System:** A ventilation fan and air inlet filter unit shall be provided for the capacitor banks to facilitate better heat dissipation, the capacitors shall be placed to ensure adequate heat dissipation by convection and radiation.
11. **Thermostat Operation:** The ventilation fan shall be operated by two thermostats set at 35°C and 55°C, respectively. (adjustable)
12. **Temperature Rise Calculations:** Temperature rise calculations shall be submitted along with the offer to prove that the temperature rise is within the temperature tolerance levels of the capacitor bank components.
13. **Temperature Rise Test:** The mounting of PFC components and their suitability shall be revalidated by a temperature rise calculation as per IEC 61921 considering min air flow available.

5.1 ENCLOSURE

5.1.1 General Requirements

1. The APFC capacitor bank shall be housed in a suitable metallic enclosure with a front door and provision for locking arrangements.
2. The dimensions of the APFC enclosures shall comply with the size of the installed components and according to technical industrial standards.
3. The APFC enclosure design shall ensure safe and reliable operation in both indoor and outdoor applications.
4. The enclosure shall be designed to give maximum reliability in service operation, inspection, and maintenance.
5. The APFC enclosure shall be of frame construction made from sheets of cold or hot rolled steel (galvanized steel optional according to EDC requirements).
6. The APFC enclosure shall comply with standard IEC 60529.
7. The APFC enclosure doors shall be provided with vertical metal support or any other suitable supporting method accepted by EDC.
8. Where necessary, ribs/reinforcements shall be used to prevent excessive wobbling, especially on

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the doors.

9. Welded structures and fabricated components shall be pretreated and then electrostatic powder coated to provide protection against environmental factors.
10. The front doors of the enclosure shall be installed using galvanized metal or stainless steel hinges (at least three hinges per door) except the type tested enclosure.
11. The doors shall be locked with a nickel-plated lock with a master key and rod to close the panel's door at three points except the type tested enclosure.
12. The enclosure door shall be provided with suitable gaskets.
13. All enclosures or partitions, including locking means and hinges for doors, shall be of mechanical strength sufficient to withstand stresses during normal service and short-circuit conditions.

5.1.2 OUTDOOR INSTALLATION

1. **Roof Enclosure:** The roof enclosure shall have a pyramidal slope on sides or back of 6 degrees with the horizontal.
2. **Enclosure Material:** The enclosure shall be made from cold or hot rolled steel with a thickness of not less than 2.5 mm before painting (or IK10 supported with certificate) (or galvanized steel optional according to EDC requirements).
3. **Painting:** The paint shall be exterior painting.
4. **Electrostatic Powder Coating:** The enclosure shall be coated with electrostatic powder with a thickness of 60-80 microns everywhere.
5. **Protection Degree:** The protection degree shall be not less than IP54.

5.1.3 INDOOR INSTALLATION

1. **Enclosure Material:** The enclosure shall be made from cold or hot rolled steel with a thickness of not less than 2 mm before painting (or galvanized steel optional according to EDC requirements).
2. **Painting:** The enclosure shall be provided with primary and finishing coat of electrostatic painting powder on the inside and outside. The external coat is preferred to be high gray color (RAL 7035).
3. **Rust Treatment:** The enclosure shall be treated against rust.
4. **Electrostatic Powder Coating:** The enclosure shall be coated with electrostatic powder with a thickness of 60-80 microns everywhere.
5. **Gaskets:** The enclosure door shall be provided with suitable gaskets.
6. **Protection Degree:** The protection degree shall be not less than IP41.
7. **Lifting Eye-Bolts:** The whole unit enclosure shall have two eye-bolts diagonals to be lifted and raising plinth.

5.2 BUS-BARS

1. **Material and Purity:** All bus-bars shall be made of rectangular, high-grade electrolytic copper with a minimum purity of 99.99% (conductivity: 57 MS/m). A certificate from a neutral and accredited body shall be provided, and the distribution company reserves the right to take a

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random sample for analysis during inspection and testing at the supplier's expense.

2. **The maximum current density of bus-bar shall be 1.5 A/mm² except the type tested panel at 45°C.**
3. **Insulation:** All bus-bars shall be insulated with colored, heat-shrinkable, non-flammable, and non-hygroscopic tubes (red, yellow, blue, and black) for the 3-phases and neutral bus-bars, respectively and the extension copper links for C.B (excluding type tested panels).
4. **Fault Current Withstand:** The bus-bar section of power factor assemblies shall withstand a minimum fault current of the system at the point of connection.
5. **Arrangement and Assembly:** Phase bus-bars shall be arranged systematically and assembled using insulators.
6. **Cross-Sectional Area:**
 - Neutral bus-bar: 100% of the CSA of the phase bus-bars.
 - Earth bus-bar: at least 25% of the CSA of the phase bus-bars.
7. **Mounting and Insulators:** Bus-bars shall be mounted on insulators made from porcelain or cast resin according to IEC standards.
8. **Short-Circuit Current Withstand:** Bus-bars and insulators shall withstand thermal and mechanical effects resulting from short-circuit currents according to IEC 61439-1.
9. **Neutral Bus-Bar Connections:** The neutral bus-bar shall be equipped with necessary compression lugs (Bimetal, Al-Cu) suitable for connection to aluminum neutral conductors and PVC-insulated cables (concerned by the contractor).

5.3 CAPACITOR BANKS

5.3.1 General Requirements

- **Modular Construction:** The APFC capacitor banks shall be of modular construction and shall be switched ON/OFF automatically in steps through the use of capacitor duty contactors.

5.3.2 Components

1. **Main Incomer Isolating ACB/MCCB**
2. **HRC Fuse (or MCCB as options):** Each stage of the capacitor bank shall also have a suitable rated HRC fuse or MCCB as options.
3. **Contactors:** Three-pole contactors for capacitor switching for each stage.
4. **Power Factor Regulator:** Microprocessor-based power factor regulator for automatic power factor correction.
5. **Capacitor Units:** Power factor improvement three-phase capacitors, arranged in a suitable number of stages.
6. **Detuning Reactors:** Detuned three-phase iron-cored series reactors for harmonic current suppression and to prevent resonance (**as options**).

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5.3.3 Design and Performance

1. **Automatic Operation:** The automatic capacitor bank should be ready for field connection with all components clearly labeled for identification.
2. **Component Coordination:** To ensure safety, reliability, and accountability of component coordination, all major capacitor bank components should be preferably from a single manufacturer.
3. **Temperature Conditions:** The capacitor bank shall be designed for trouble-free service under arduous temperature conditions.

5.4 CAPACITOR UNITS

- The capacitor units specification shall comply with EDMS 21-301

5.5 Switching and protective devices and connections

5.5.1 Transient Overcurrent Withstand

- The switching and protective devices and connections shall be capable of withstanding electrodynamic and thermal stresses caused by transient overcurrent of high amplitude and frequency which occurs during switching on.

5.5.2 Cable and Circuitry Selection

- The connecting cables and circuitry shall be selected to handle 1.5 times the rated capacitor current permanently.

5.5.3 Protection Requirements

1. The capacitor bank shall be protected by a suitable rated MCCB/ACB at the incomer.
2. HRC Fuses (MCCBs as option) shall be used for short-circuit protection.
3. HRC Fuses do not generally provide suitable overcurrent protection.

5.5.4 Circuit Breaker Specifications

1. The MCCB OR ACB shall comply with EDMS 19-303
2. The current rating of the MCCB shall be at least 1.5 times the full load current of the capacitor bank.

5.5.5 Fuse and its Base Specifications

1. The HRC Fuses shall comply with EDMS 19-300
2. The fuses base shall comply with EDMS 19-301
3. The HRC fuse rating shall be (1.5) times the nominal capacitor current.

5.5.6 CAPACITOR DUTY CONTACTORS

1. Capacitor duty contactors is Specially designed for switching capacitors and provided with damping resistors which limit the value of inrush current to a save value.
2. The Capacitor duty contactors shall isolate all three supply phases to the capacitor on switch off.
3. The capacitor duty contactors are rated based on nominal reactive power.
4. The rated voltage of Capacitor duty contactors shall be equal to or higher than the maximum network voltage with the power factor correction installation.
5. The Capacitor duty contactors shall be designed for continuous current (including harmonics) which can pass the power factor correction installation and / or filter at maximum frequency and

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extreme tolerance of the components, especially capacitors and reactors.

5.5.6.1 Contactors specifications (stage switching)

1. The switching contactor specifications shall comply with IEC 60947-4-1
2. The Capacitor duty contactors shall be three poles type and shall be employed for switching ON and switching OFF operations in capacitor banks.

3. General Technical Data

Table (4)

Item	Unit	SPEC
Rated operation voltage (Ue)	(V)	400/440/480/525 as option
Operating voltage coil	(V)	220 VAC $\pm$ 10%
Frequency	(HZ)	50
Rated Insulation voltage (Ui)	V	690VAC
utilization category	Class	AC 6b
Minimum auxiliary contacts	No.	(1 NO + 1 NC)
Number of operations	Op	240 per hour
Electrical life (operation under capacitive switching conditions)	Op	250000
Degree of protection	IP	IP20
Storage temperature	°C	-60°C:80°C
operation temperature	°C	$\geq$ 55°C
Humidity	%	up to 95% (non-condensing)
Rated Impulse Withstand Voltage	KV	8 KV

5.6 FILTER, DETUNING REACTORS ASSOCIATED WITH CAPACITORS (as option)**5.6.1 General**

1. Filter reactors shall be used in series with the capacitors for cases where resonance is most likely to occur, this arrangement will ensure the reduction of harmonic distortion and avoidance of series / parallel resonance that can lead to:
 - Overloading of capacitors
 - Overloading of transformer and transmission equipment
 - Interference with metering and control system
 - Resonance elevation i.e. amplification of harmonics
 - Voltage distortion
2. Capacitors shall be positioned so as to avoid the occurrence of resonance, the same value of KVAR installed at high voltage rather than low voltage can eliminate a resonant condition.
3. This clause describes the requirements for reactors designed to be used in association with capacitors.
4. On main supplies with high level of harmonic pollution, it is recommended to use capacitor banks with detuned reactors which have the following functions:
 - Increasing the capacitor impedance in relation to the harmonic currents.
 - Shifting the parallel resonance frequency of the source and the capacitor to below the main frequencies of the harmonic currents that are causing interference.

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- Filter reactors connected in series or in parallel with capacitors to reduce or block harmonics or control signals (ripple signals).

5.6.2 Detuned filtering (PFC-DF)

1. (Depending on the distribution company requirement: according to load analysis)
2. Capacitors with detuned filtering technique shall be employed to correct power factor while avoiding the risk of resonance condition.
3. This shall be performed by shifting the resonance frequency to lower values where no harmonic currents are present, by introducing a filter reactor in series with the capacitors, such that the capacitor/reactor combination is inductive at the dangerous frequencies but capacitive at fundamental frequency.
4. The circuit should be tuned such that the series resonant frequency should be below the lowest harmonic order expected to be present in the electrical network.
5. The detuning reactors shall be connected in series with each capacitor stage and shall be of iron cored type.
6. The detuned reactor should be equipped with thermal switch inside the winding which should cut-off the respective contactor in case of over-temperature.
7. The reactors shall be selected taking into account the inductance value of the feeding line so as to get the desired tuning frequency and the current capability high enough for the harmonic current absorption that can be expected.
8. Filter reactors shall be designed to give an inductive voltage drop up to 10% of the system nominal voltage, thus protecting the capacitors from harmonics.

5.6.3 DETUNING REACTORS SPECIFICATIONS

1. The detuned reactors should be suitable to capacitor banks according to design.
2. The detuning reactor shall be manufactured in full compliance with and tested to the requirements of IEC 60076-6 Reactors.
3. The reactor insulation shall be class "F" or above
4. The maximum temperature of the reactor at maximum continuous RMS shall be no higher than 145 °C at a 50°C ambient.
5. If the lowest harmonic order expected to be present in the electrical network is likely to the 150 HZ (3rd harmonics) then the series detuning reactors should be tuned for 135HZ 2.7 tuning order) with a relative impedance 14%.
6. Tolerance of reactors shall not exceed $\pm 3\%$ of rated value.
7. IEC 60289, IEC 60076-6, VDE 0532
8. Temperature class F 155°C (temperature rise limit at coils on air) the classification of insulations is that given in clause 2 of IEC 60085
9. Temperature protection 130 °C $\pm$ 5 %
10. The reactor shall be securely fastened and installed in a separate compartment of the power factor correction board, if necessary rubber pad shall be used to reduce noise.

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Table (5)

Item	Unit	SPEC
Standard	IEC	IEC60076-6
Rated Voltage U_r	(V)	400
Rated frequency	HZ	50
Tuning order	%	2.7
Tolerance in inductance	%	3 %
Insulation class	class	F(H OPTION)
Blocking Factor p%	%	5.67-7%-14% According to calculation
Basic Insulation Level	KV	0.525/2.5/8 KV- AC
Power losses	W/ KVAR	3.5 – 6 W/ KVAR
Windings material	type	CU, (AL as option)
Reactors type	type	dry iron core
Cooling method	type	AIR
Flux density	tesla	<0.8 tesla
Dielectric strength	KV	3KV /1min to IEC60076-3
Noise level	dB	acc to IEC required

5.6.4 Harmonic filtering

1. For elimination of harmonics from the system, real LC filters shall be installed
2. The bank shall be divided into several series filters, each tuned to the harmonic frequency that is supposed to be eliminated.
3. The LC circuit, tuned to each harmonic order shall be installed with the nonlinear load that will function to absorb the harmonics, thus avoiding their flow in the distribution network

5.6.5 Marking

1. reactor shall be having rating name plate of weather proof materials, fitted in a visible position, showing in all cases the appropriate items indicated below, the entries on the name plate shall be indelibly marked.
2. Type of reactor - Outdoor/indoor application – IEC – Manufacturer`s name - Manufacturer`s serial number – year of manufacture – insulation level – No. of phases – rated frequency – highest voltage – rated continuous current – rated thermal short circuit current and duration-thermal class (for dry type) – total mass – type of cooling.

5.7 POWER FACTOR CONTROLLER (Regulator)**5.7.1 General Requirements**

1. The PFC controller is the brain of the APFC system, switching ON/OFF steps based on the KVAR required to maintain the power factor close to unity.
2. The regulator shall automatically and/or manually step-in or step-out various capacitors to correct the power factor to the required pre-set level.

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3. The regulator shall be insensitive to harmonics and screened from stray electric and magnetic interference.
4. The regulator shall be housed in a case suitable for panel mounting.
5. Operating temperature shall be -5°C to $+70^{\circ}\text{C}$.
6. The regulator shall be suitable for 1A or 5A current input.
7. The regulator (controller) has IP 42 as minimum.

5.7.2 Technical Requirements

1. The regulator shall be 4, 6, 8, 10, 12, or 14 steps type as specified in the drawing and/or Bill of Quantities.
2. No voltage trip and zero current trip shall be provided.
3. For a voltage loss of longer than 15ms or current loss of longer than 3s, all capacitor stages connected shall be switched off until discharged, after voltage or current is stored the regulator switches the required steps on.
4. The regulator shall have an LCD display (2x16 characters) to indicate harmonics, current, voltage, frequency, power factor, active and reactive powers.
5. The controller shall have LED/LCD display. Parameter in display should be cyclic with suitable time delay.
6. All capacitor units (except fixed unit) should be controlled by a APFC controller, responsible for monitoring the reactive and active components of the network through the current and the voltage path.
7. The capacitor stages shall be switched in and out depending on the deviations of the actual power factor from the set value as calculated by the controller.
8. The controller shall sense current from a single phase and voltage from two phases to compute the average 3-phase power factor and accordingly compensate by switching in necessary capacitor stages.

5.7.3 Functional Requirements

1. The regulator shall automatically program and control the capacitor bank.
2. The regulator shall sense the reactive current requirement and switch on/off the required stages of the capacitor bank.
3. The regulator shall maintain a targeted power factor within 0.95 lagging and unity.
4. The regulator shall be microprocessor-based.
5. The regulator shall detect any stage size by automatic recognition.
6. The regulator shall have a minimum time delay of 120 seconds for switching on a capacitor.
7. The regulator shall be panel-mounted and easily programmable.
8. The regulator shall conform to safety guidelines as per IEC 61010-1-2001.
9. The regulator shall have programmable dual cos phi.
10. The display unit shall be easily visible, preferably without opening the APFC cabinet door/cover.
11. The operating voltage shall be 380 volt / 220 volt so the controller can be connected 3 phases or 1 phase.
12. It can indicate the odd harmonics for voltage (as option).
13. It can switch off the stages when voltage harmonics exceeds the limits with IEEE519(as option).

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14. It can display actual power factor and target power factor, active current, reactive current, the number of stages in service.
15. It can store the data in spite of the supply with switch off.
16. It can be operated in different programs.
17. It shall be flush mounted for indoor panels, and it shall be surface mounted for outdoor.
18. Regulator (controller) should have a possibility for downloading data through PC (as option).
19. The voltage on the capacitor terminals may be particularly high at times of light load conditions. In such cases, some or all of the capacitors shall be switched out of circuit in order to prevent overstressing of the capacitors and under voltage increase in the distribution system.
20. The controller shall be easily programmable from front panel itself.
21. The automatic power factor controller shall be manufactured in full compliance with and tested to the requirements of IEC 61326-1 (ELECTRICAL EQUIPMENT FOR MEASUREMENT, CONTROL AND LABORATORY USE – EMC REQUIREMENTS – PART 1 : GENERAL REQUIREMENTS)
22. The capacitor bank and controller shall ensure that after the loss by fault of any one stage, it shall continue to operate automatically and shall follow rotational switching.
23. The threshold value for the operating temperature and system harmonics shall be programmable and the regulator shall be able to switch off the connected capacitor stages if the actual values exceed these thresholds.
24. Dual Cos Phi – the controller should have programmable dual cos phi to differentiate the need in compensation (Cos Phi) when the operating condition changes (as option).

5.7.4 SETTING AND DISPLAY

5.7.4.1 The power factor controller / regulator shall be incorporated the following setting:

1. It has the capability to adjust setting time of contact between stages
2. Recognize the connection of CT and be able to automatically adjust itself to the phase angle difference.
3. Regulator shall have the capability to automatically search and set the C/K setting, it shall be also possible to program the C/K setting manually.
4. C/K inductive setting and display
5. C/K capacitive setting and display
6. Target power factor setting and display (0.8 inductive-1-0.8 capacitive)
7. Operation delay time setting and display (5 s to 300s)

5.7.4.2 The power factor controller / regulator should allow the following readings:

1. Automatic initialization and stage rating detection
2. Any step sequence detection (user definable step sequence)
3. Measurement of capacitance per stage
4. Cap bank over load current ration
5. THD Voltage
6. 4 quadrant operation
7. Record of the Max temp internal of the capacitor bank since reset
8. System Voltage (V AC)

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9. Temperature (°C)
10. Real time Cos phi
11. KVAR value to target Cos Phi

5.7.4.3 The controller shall initiate alarms and warnings in the following events.

1. Temperature limit is exceeded
2. Insufficient capacitor output
3. Overload current ratio limit is exceeded
4. THDV limit is exceeded (as option)
5. Low power factor / under compensation
6. Over compensation
7. Over current
8. Capacitor step defective
9. Over voltage
10. Total voltage harmonic distortion THDV

5.7.5 Mode of operation

5.7.5.1 Manual operation mode

- The capacitor steps can be switched in step by step and switched out in reverse step by step manually, and the step that is being switched on shall be displayed.

5.7.5.2 automatic operation mode

- The regulator shall switch on or switch out the required capacitors without undue switching on and switching off at any particular step to reach the nominal target power factor.

5.7.5.3 OFF mode

- All equipment shall be switched off and de-energized.

5.8 CURRENT TRANSFORMERS

- The current transformer shall comply with EDMS 17-300 and shall be specified according to KVAR panel rating

5.9 NAME PLATE, SAFETY INSTRUCTORS

1. Name plate for each panel shall include the following data:
 - The name of Distribution Company.
 - The rated of the bank in KVAR.
 - The rated voltage and rated frequency.
 - The date of production.
 - Three lamp led for voltages
2. Plastic sheet should be installed in a visible place at the front of the panel and shall include the following safety instructions with distinguished color, with suitable size (electrical hazard, don't touch the capacitor unit and all parts inside the panel unless safety time not less than 5 minute may be by English or Arabic language).

5.10 VENTILATION

1. The enclosure shall be air forced ventilation according to the following:
2. Install two thermostats on the top section of the capacitor bank away from the dissipating devices:
3. First thermostat: to switch on the fans, when the internal temperature exceeds 35 °C

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4. Second thermostat: to trip the capacitor bank if the internal temperature exceeds 55 °C
5. The selection of fans shall be based on several parameters:
 - IP level of the panel
 - Total watt loss including all components in the panel
 - Total volume of the enclosure
 - Outside ambient temperature
6. The fins of the ventilation fans should be made from metal.
7. For better air circulation, a minimum clearance of 30 mm between capacitor units and panel enclosure need to be kept
8. Also maintain a 30mm gap above the top of the capacitor.
9. Desired internal ambient temperature

Table (6)

Nominal voltage	VAC	220-240
Frequency	Hz	50
Mn. ambient temperature	C°	-10
Max. ambient temperature	C°	55
Min. air flow	m ³ /h	≥ 40 for 50 k VAR ≥ 60 for 75 k VAR ≥ 80 for 100 k VAR ≥ 120 for 150 k VAR ≥ 160 for 200 k VAR ≥ 200 for 250 k VAR ≥ 240 for 300 k VAR ≥ 280 for 350 k VAR ≥ 320 for 400 k VAR ≥ 360 for 450 k VAR ≥ 400 for 500 k VAR
Connection line		2 flat plugs
Bearing		Sleeve or ball bearing system
Life time at 40C°		37500 or more
Housing material		Metal
Material of impeller		Metal

10. This rating above without reactors

5.11 COMPONENTS RATING CALCULATIONS

1. The manufacturer shall comply with any calculations submitted to him by table 7 & table 8 and by...EDC without detuning reactors such as
 - CB rating calculation
 - Fuse rating calculation
 - Contactor rating calculation
 - Cable size calculation
 - Current transformer

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2. Rating of HRC fuse and Contactor, C.S.A of Conductor for individual capacitor:

Table (7)

CAPACITOR RATING (KVAR)	Rating current(A)	C.S.A of conductor mm2,copper	Fuse rating (A) (K.VAR*1.443*1.5A)	Contactor rating (A)
2.5	3.6	1.5	10	≥10
5	7.2	2.5	16	≥10
10	14.4	4	25	≥20
12.5	18	6	32	≥25
15	21.6	10	40	≥30
20	28.86	10	50	≥39
25	36	16	63	≥49
50 / (2*25)	72.15	50	125	≥98

3. Rating of C.B for capacitor panel:

Table (8)

CAPACITOR RATING (KVAR)	Rating current(A)	C.B RATING (A)
		(K.VAR*1.443*1.5 A)
2.5	3.6	16
5	7.2	16
7.5	10.8	16
10	14.4	32
12.5	18	40
15	21.6	40
20	28.86	63
25	36	63
30	43.3	80
35	50.5	80
40	57.72	100
45	65	100
50	72.15	125

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75	108.2	200
100	144.3	250
150	216.45	400
200	288.6	630
250	360.75	630
300	432.9	800
350	505	800
400	577.2	1000
450	649.35	1000
500	721.5	1250
٦٠٠	٨٦٥,٨	١٦٠٠

5.12 INTERNAL WIRING

1. All small wiring shall have a cross sectional area of not less than 2.5 mm and shall be 0.6/1 KV, insulated with class 5 PVC copper conductors.
2. All small wiring shall be suitably terminated and fitted identification ferrules and marked with circuit numbers in accordance with approved standard.
3. The trip circuit shall have an additional ferrule colored red and marking trip.
4. All terminal blocks shall be from an approved design and all exposed terminals shall be enclosed by detachable covers.
5. The trip circuit cables shall be colored black and the instrument transformer cables (CTs) colored with their respective phase colors.
6. The wiring of the auxiliary circuits shall be segregated from the main circuit by heat resistant tubes made of insulating material covered by earthed metallic partitions.
7. Terminal blocks, small wiring terminations, ferrules and wire numbering and coloring shall be subjected to the approval of the purchaser.

5.13 TESTING AND INSPECTION

1. Tests should be performed at the manufacturer's works in accordance with the relevant latest IEC standards.
2. The tests shall include:

5.13.1 ROUTINE TESTS

1. All tests shall be carried out in the presence of engineer in charge, at such times as he may reasonably require.
2. All samples used for testing shall be to the contractor's expense and shall not affect the quantities to be supplied under this contract.
3. All instrument used for testing purpose, shall be calibrated by an approved authority.
4. The cost of all tests shall be include in the contract price and shall not be quoted for separately.

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5. Routine tests of the capacitor bank should be carried out according to latest relevant IEC standards by the manufacturer; a representative of --EDC will attend these tests before acceptance.
6. The --EDC have the right to add any test during delivery to check the quality of the product.
 - Visual inspection.
 - Mechanical operating tests.
 - Electrical operating tests.
 - Dielectric test.

5.13.1.1 Routine tests for capacitor bank

1. All routine tests on the assembled capacitor bank shall be carried out in accordance with IEC61921 and IEC61439-1 as per the relevant clauses mentioned against each test.
2. Routine tests shall be carried out by the manufacturer on every APFC PANELS before delivery.
 - Inspection for conformity with specifications.
 - Inspection of the assembly including inspection of wiring and if necessary an electrical operation test.
 - Inspection, including checking of clearance, dimensions.
 - Dielectric test shall be as per IEC61439
 - Checking of protective measures and of the electrical continuity of the protective circuit
 - Verification of insulation resistance
 - During inspection, routine test shall be carried out on random samples selected by Distribution Company's representative (As distribution company requirements).

5.13.1.2 Routine tests for capacitor unit

- According to EDMS 21-301-1

5.13.1.3 Contactor tests

- Design and visual checks.
- Dielectric test on main circuit
- Tests on auxiliary and control circuit

5.13.1.4 Reactor unit tests

- Measuring winding resistance.
- Measuring od impedance at rated continuous current.
- Measuring of losses at ambient temperature.
- Winding of over voltage test.

5.13.2 TYPE TESTS

1. Type test certificate should be issued by an independent approved test laboratory by EEHC.
2. All equipment shall be successfully type tested as EEHC approved laboratory in accordance to latest relevant standard for the following,
3. Type testing certificate and reports shall be submitted from independent certification bodies and must include the following:

5.13.2.1 Type tests for Capacitor

- All type tests shall be carried out in accordance to relevant IEC
 - Thermal stability test
 - Measurement of ($\tan \delta$) of a capacitor.
 - AC test between terminals and container.

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- Lightning impulse voltage test between terminals and container.
- Over voltage test.
- Short circuit discharge test.
- Disconnecting test on internal fuses.
- Test of an external fuse in a combination with a capacitor.

5.13.2.2 Type tests on MCCB'S, Reactors, Current Transformers

- Complete type test certificates for all major components of the panel shall be submitted as per the relevant IEC standards along with the tender.

5.13.2.3 Type tests of contactor

- All type tests shall be carried out in accordance to relevant IEC60947-4-1
 - Mechanical endurance test.
 - Making and breaking and reversibility test.
 - Overload current withstand tests
 - Short-circuit current making and breaking tests.
 - Verification of operating limits and characteristics of overload relays.
 - Change-over ability and reversibility, where applicable.
 - Test of the striker mechanism.

5.13.2.4 Type tests of Reactor

- All type tests shall be carried out in accordance to relevant 60076-1&6
 - Temperature rise test at rated continuous current.
 - Lightning impulse test for current-limiting reactor.
 - Measuring of power consumption of fans and oil pumps if any.

5.13.3 SITE TESTS

1. The contractor is responsible for submitting all contract works to site inspection by the engineer, before site tests are commenced.
2. Before commissioning, the contractor shall depute an experienced and qualified testing engineer from the manufacturer's works to carry out the following tests on the equipment and such other tests that may be considered necessary by the Purchaser.
3. The Site Acceptance Test (SAT) format for the capacitor bank shall be forwarded to the purchaser prior to the SAT.
4. The contractor's test engineer shall complete all pre-commissioning test, commission all plant and equipment supplied by him and hand over the entire contract works to the Purchaser in good shape.
5. All the charges connected with the pre-commissioning tests of the equipment shall be included in the tender price.
6. The contractor's testing engineer shall carry out all commissioning tests in co-operation with and to the satisfaction of the Purchaser's engineer who will take part in all these tests.
7. The manufacturer shall provide all test equipment require for different test purpose at site
8. The following test/inspections at site shall be carried out:

5.13.4 Mechanical Tests

1. Visual inspection to verify degree of protection creepage and clearance distances.
2. All conductors and cables are checked for proper routing and all devices for proper mounting.
3. Check effectiveness of all mechanical devices, e.g. handles, locks, interlocks, operating devices,

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etc.

4. Check panel conformity to drawing and engineer's requirements.

- Checking of all mounting plates/fasteners.
- Checking of dimensions and components as per drawings.
- Electrical circuits fasteners tightness / surfaces area contact
- Crimping and ferrules as per drawing.
- Labels / Identification / Nameplate.
- All doors checking, safety and accessibility
- APFC cabinet surfaces finish / smoothness

5.13.5 Electrical Tests:

1. Insulation resistance test shall be carried out at all main circuits through to final terminal.
2. Insulation resistance shall exceed 10 MΩ.
3. Record all measurements.
4. Function test of all circuit breakers switches, contacts, etc. and every circuit to verify correct operation.
 - Insulation resistance tests between phases and earth and between neutral and earth.
 - Operational test on components.
 - Switching ON/OFF of capacitor bank on various KVAR requirement.
 - Checking of display parameters
 - Switching on/off logic verification
 - Data communication through serial / optical port
 - Verification of data / reports / functions in base computer software.

5.13.6 Final inspection

1. Prior to energizing the capacitor bank the following checks shall have carried out at site:
 - Operate the equipment through all design functions, including remote operation, actuation of alarm and indicating devices, mechanical and electrical tripping and closing and operation of the protective devices.
 - Insulation resistance measurements on the buses, phase to phase and phase to ground, with all breakers in the fully connected position and contacts open.
 - Control circuit insulation resistance to ground.
 - Inspection all relays and protective devices, and verify setting in accordance with the manufacturer's instructions. Inspect current transformers and relays for correct polarity of connections and the installation of jumpers on unused current transformer circuits.
 - Manually close and trip each breaker checking and adjusting the main contact alignment and wiring action in accordance with the manufacturer's instructions.
 - Test protective relay operation for incomer air circuit breakers.
8. With the capacitor bank in operation, measurement of the power factor and system harmonics shall be carried out after commissioning of equipment.

5.14 DRAWINGS AND INFORMATIN

- The manufacturer shall furnish the following drawings and documents placed inside the drawing pocket of the equipment supplied:
 - Dimensioned drawing showing outline of the capacitor bank

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- Single line diagram showing all the major electrical components
- Protection and control schematics of the capacitor bank
- Details of cable terminations and fittings.
- Technical manual giving installation, operation and maintenance instructions.

5.15 GROUNDING

1. The capacitor bank should be provided with suitable ground terminal.
2. All bank and enclosure shall be connected to a 40*6 mm earthing copper busbar.

5.16 HANDLING

1. A shock control shall be placed on the package of the capacitor bank. If the bank falls, this control will be broken.
2. The equipment shall be provided with lifting rings, or provided with provision for using a Forklift truck for handling.
3. For Installation, a 250 mm minimum space should be provided at the back of the bank in order to have an optimal position of the bank for good ventilation.

5.17 CERTIFICATION

- All main component such as: Capacitor C.B, contactors, (fuse link & fuse base) and... should be approved by EEHC.

5.18 WARRANTY/DISCLAIMERS

1. The Panel of capacitor bank assembly shall be warranted by the manufacturer to be free from defects for a period of 30 months from factory shipment or a period of two years after the unit is energized, whichever occurs later.
2. In case of any component defected during the warranty period, the warranty period will be restarted from the replacement date of this defected component.
3. The manufacture shall comply with any additional distribution company requirements, (as optional).
4. Spare parts shall be supplied according to the distribution company requirement and their prices will be fixed at the envelope opening during the warranty period.

5.19 MARKING

- The following data should be engraved or laser printed on a nameplate riveted on the Enclosure and located in a visible and legible place showing the following information:
 - Manufacturer's name
 - Type/Model
 - Serial Number and year of Manufacturing
 - Distribution company name
 - Order number
 - Rated total output in KVAR
 - Number of capacitor stages with capacity for each stage.
 - Rated voltage and rated insulation level
 - Rated frequency
 - Guarantee period
 - Date of dispatch

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- Maximum power consumption pf APFC panel.

5.20 SUBMITTALS

- Vendor shall provide the following with the quotation:
 - Clause by clause compliance with the specification.
 - All necessary drawings including the scheme drawing of the power capacitor bank, dimensions, single line diagram of (control circuit, ...etc.)
 - Technical catalogues containing the technical data for all offered components of the unit should be submitted.
 - Installation and maintenance instructions.
 - Copy of type test report for capacitor, contactor, MCCB and reactor.
 - Submitted a valid approved letter from the Egyptian Electricity Holding Company if any.
- Vendor shall provide the following after signing of purchase order:
 - Details of manufacturing and testing schedules
 - Routine test reports (including tightness test)
- Vendor shall provide the following with delivering:
 - Operation (user) manual.
 - Handles, Master Keys, Locks..... for operation and locking.

5.21 AFTER SALE SERVICES

- The tenderer should have a service center in Egypt covers all its deliverables.
- EDC reserves its right to claim a free after sale service all over the warranty period guaranteed by the tenderer that covers all its deliverables, after sale services should include and not limited to the following:
 - Technical support.
 - Maintenance.
 - Original Spare parts.
 - Submittals.
- All the services, spare parts, materials and accessories will be needed during the after-sale maintenance should be guaranteed by the tenderer immediately and wherever required (Warehouse – Site – Lab.etc.)

5.22 GUARANTEE

- The supplier guarantees the panel against all defects arising out of faulty design or workmanship, or of defective material for period of 24 months from date of delivery or 18 months from date of operation.
- The good quality and manufacturing of the materials should be guaranteed.
- Any parts which on account of poor quality of the materials, manufacturing defects or poor assembly, should be replaced or repaired in the shortest time possible and free of charge.

4. TECHNICAL DATA SCHEDULE

- The guarantee schedules should be submitted and the tenderer should fill in the attached guarantee tables.
- EDC reserves the right to reject any tender not accompanied with clear and complete guarantee tables.

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GUARANTEE TABLES

Table (1)
(POWER FACTOR REGULATOR)

No.	ITEM	EEHC Specs.	Supplier Specs.	Remarks
1	Type /model NO			
2	Country of manufacture			
3	Type of microprocessor based			
4	Number of steps			
5	Rated voltage			
6	ambient temperature °C			
7	Modes of operation			
8	Setting			
9	Display			
10	Alarms			
11	Operating Temperature °C			
12	Storage Temperature °C			
13	Frequency			
14	Rated Current A			

Table (2)
(DETUNED REACTOR)

No.	ITEM	EEHC Specs.	Supplier Specs.	Remarks
1	Inductance			
2	Country of manufacture			
3	Type /model NO			
4	Winding material			
5	Rated voltage			
6	operating temperature °C			
7	ambient temperature °C			

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Table (3)
(SWITCHING CONTACTORS)

No.	ITEM	EEHC Specs.	Supplier Specs.	Remarks
1	Type /model NO			
2	Country of manufacture			
3	Utilization category			
4	Number of steps			
5	Rated voltage			
6	Ambient temperature °C			
7	Rated voltage for operating coil			
8	operating temperature °C			

Table (4)
(VENTILATION FANS)

No.	ITEM	EEHC Specs.	Supplier Specs.	Remarks
1	Type /model NO			
2	Country of manufacture			
3	Diameter and dimensions			
4	Air exchange			
5	Rated voltage			
6	Ambient temperature °C			
7	Rated voltage for operating coil			
8	NO of auxiliary contacts			
9	KVAR Rating			
10	operating temperature °C			

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Table (5)

(General)

No	Item	Vendor Spcls.
General		
1	Rated (Maximum operating) voltage	V
2	Nominal operating (Service) voltage	V
3	Power frequency R.M.S. withstand voltage (1 min)	V
4	Impulse withstand voltage (1.2/50) μ s peak	V
5	Rated frequency	HZ
6	Color	
7	Auxiliary voltage	V
8	IP Class	
9	Copper purity	%
10	Copper Conductivity	MS/m
Environmental conditions		
1	Ambient temperature	°C
2	Average relative humidity	°C
3	Maximum altitude	m
Design and construction requirements		
MCCB /ACB		
1	Rated Voltage	KV
2	Rated normal current at 40&45&50 C°	/...../ A
3	Short-time withstand current	KA
4	Making capacity	KA
5	Mechanical endurance	
6	Electrical endurance	